

COLE BROOKS

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PROFESSIONAL PROFILE

Fullstack and embedded systems engineer with 4+ years of experience building FDA Class II medical device software, secure cloud infrastructure, and production IoT systems. Experienced owning major features across firmware, Python applications, and AWS services, delivering compliant, scalable solutions in regulated environments.

EDUCATION

Montana State University
B.S. in Computer Science

November 2022

EXPERIENCE

Titin KM Biomedical

Software Engineering Intern

October 2021 - November 2022

Bozeman, MT

- Contributed to proof of concept software for prototype data driven medical rehabilitation device used to secure over one million dollars in seed funding.

Titin KM Biomedical

Junior Software Engineer

November 2022 - January 2025

Bozeman, MT

- Owned broad features in a cross platform Python/Kivy application used by clinicians to collect and analyze biomechanical data; reduced data collection time by approximately 3x and enabled clinicians to perform new data driven assessments.
- Designed and implemented the application's MVVM architecture to improve structure, maintainability, and testability.
- Implemented C++ firmware modules for real time biomechanical measurement, contributing to system data accuracy and reliability.
- Collaborated with electrical engineering to define wiring harness specifications with appropriate strain relief and EMI mitigation, contributing to overall system reliability and FCC compliance.
- Collaborated with mechanical, electrical, and regulatory teams during development and verification of an FDA Class II medical device.
- Developed GStreamer/WebRTC based screen mirroring tool used by sales teams for live demonstrations.

Titin KM Biomedical

Software Engineer

January 2025 - Present

Bozeman, MT

- Designed and deployed compliant FastAPI & AWS backend, supporting a growing fleet of deployed medical devices with daily device-to-cloud telemetry and enabling remote diagnostics and firmware updates.
- Implemented GitLab CI/CD pipelines, coding standards, and embedded test automation, reducing release cycle time by 20% and improving reliability of deployments.
- Designed zero touch IoT provisioning system (Cloud-init + AWS IoT Core/Greengrass), cutting setup time from 60+ minutes to 15 minutes and eliminating manual configuration errors.
- Developed and integrated a Python unittest test suite, improving automated test coverage and catching regressions early in development.
- Guided junior engineers and interns through design discussions, debugging, and code reviews.

TECHNICAL STRENGTHS

Programming Languages	Python, C, C++, Bash
Frontend	Kivy, Vue.js
Backend	FastAPI, PostgreSQL, SQLAlchemy
Cloud	AWS (Cloudformation, Lambda, IoT Core, Greengrass), Docker, GitLab CI/CD
Electrical	Microcontrollers, Hardware Debugging, Wiring Harness Validation
Regulatory	FDA Class II, HIPAA

ADDITIONAL ACTIVITIES

Contribution to Kivy - Updated the Kivy project's Emacs major mode to improve syntax highlighting.

Contribution to GhettoVCB - GhettoVCB is an open source VMware virtual machine backup tool. Provided a bugfix to address broken snapshots of thick provisioned, zeroed disks in VMWare.